

Solve

BRAC University (Department of Computer Science and Engineering)
CSE 320 (Data Communication) | Summer 2023

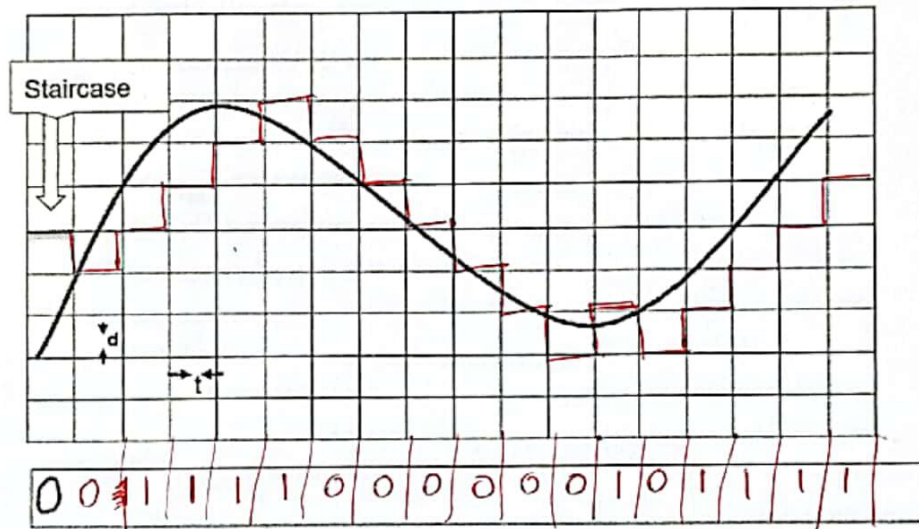
Quiz 4 (Set A)

Full Marks: 15

Duration: 30 minutes

Name:	ID:	Section:
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1. Draw the staircase in the following graph and generate the digital data from the given analog signal using the Delta Modulation (DM) technique. You have to answer this question in the question paper only. [4 marks]



2. Suppose, a voice channel occupies a bandwidth of 5 kHz. We need to multiplex 10 voice channels with guard bands of 500 Hz using FDM. Calculate the required bandwidth of the link. [1 marks]

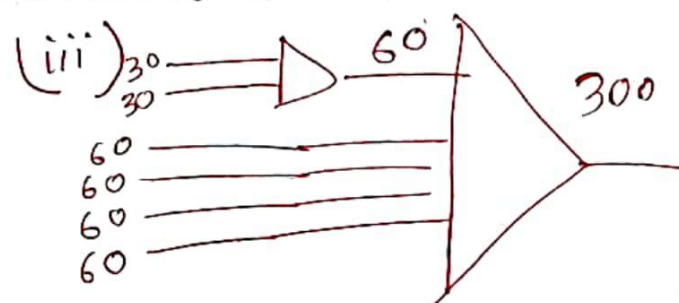
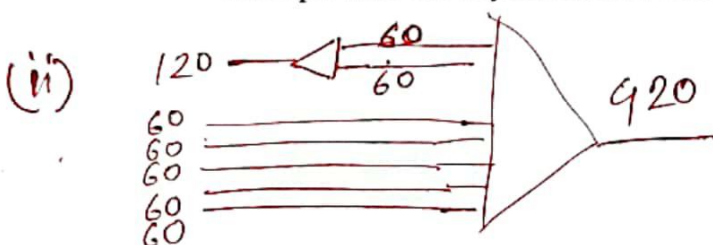
Ans: $5 \times 1000 \times 10 + 9 \times 500 = 54500 \text{ Hz}$

3. (i) Suppose, you have 6 voice channels, each of 60 Mbps. You have to use synchronous TDM to multiplex these channels. If 2 characters at a time are multiplexed: [6 marks]

- What is the size of an output frame in bits? $(2 \times 8) \times 6 = 96 \text{ bits}$
- What is the output frame rate? $\rightarrow 3.75 \times 10^6 \text{ fps}$
- What is the duration of an output frame? $\rightarrow 2.67 \times 10^{-7} \text{ s}$
- What is the output data rate? $\rightarrow 360 \text{ Mbps}$
- What is the duration of an input bit? $\rightarrow 1.67 \times 10^{-8} \text{ s}$
- What is the duration of an output bit? $\rightarrow 2.78 \times 10^{-9} \text{ s}$

- (ii) In the above scenario of question 3(i), suddenly one of the 6 channels started sending data at 120 Mbps. What will be your solution? Justify your answer with a diagram. [2 marks]

- (ii) In the above scenario of question 3(i), suddenly two of the 6 channels started sending data at 30 Mbps. What will be your solution? Justify your answer with a diagram. [2 marks]



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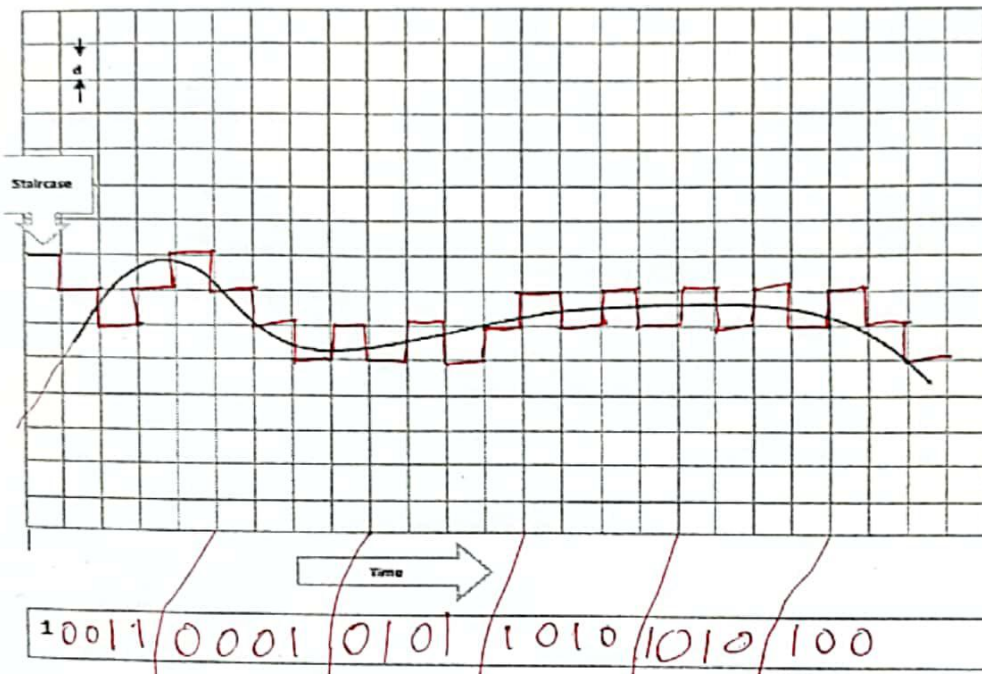
Quiz 4 (Set B)

Full Marks: 15

Duration: 30 minutes

Name:	ID:	Section:
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1. Draw the staircase in the following graph and generate the digital data from the given analog signal using the Delta Modulation (DM) technique. You have to answer this question in the question paper only. [4 marks]



2. Suppose, a voice channel occupies a bandwidth of 10 kHz. We need to multiplex 8 voice channels with guard bands of 600 Hz using FDM. Calculate the required bandwidth of the link. [1 marks]

Ans: $10 \times 10^3 \times 8 + 600 \times 7 = 84260 \text{ Hz}$

3. (i) Suppose, you have 5 voice channels, each of 50 Mbps. You have to use synchronous TDM to multiplex these channels. If 2 bytes at a time are multiplexed: [6 marks]

- What is the size of an output frame in bits? $2 \times 8 \times 5 = 80 \text{ bits}$
- What is the output-frame rate? $3.125 \times 10^6 \text{ fps}$
- What is the duration of an output frame? $3.2 \times 10^{-7} \text{ s}$
- What is the output data rate? 250 Mbps
- What is the duration of an input bit? $\frac{1}{50 \times 10^6} = 2 \times 10^{-8} \text{ s}$
- What is the duration of an output bit? $\frac{1}{250 \times 10^6} = 4 \times 10^{-9} \text{ s}$

- (ii) In the above scenario of question 3(i), suddenly one of the 5 channels started sending data at 48 Mbps. What will be your solution? Justify your answer with a diagram. [2 marks]

